Virtual University of Pakistan

STA-100

General Mathematics & Biostatistics

| Final term past papers |2019

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M.C.Q

1. **M.C.Qs** mostly from **lesson** 10,11,12,41,42,43,44,45
2. If A={2,4,8,10,12,14,16} ; B={12,13,14,15,16} then A∩B =_____**A∩B={12,14,16}**
3. Let Z_1= 2-3*i* , Z_2= -2+3*i* then $Z_1 - Z_2$ will be____
4. Let $f(x) = 5x^3 + 3x - 9$ & $g(x) = x^2$ then $f.g(x)$ =_____
5. The quadratic formulae for $ax^2 + bx + c = 0$ is_____
6. To multiply a Matrix by a scalar we multiply each ____ in the matrix by that scalar.
7. Date obtained by ministry of food is example of _____**Collection Secondary data**
8. During the current population census, what types of data was collected **COLLECTION THROUGH ENUMERATORS:**
9. We take ____ on y-axis to construct a frequency polygon.
10. The formulae of mid range is_____
11. ____ is not a measure of dispersion?
12. What will be the value of total amount of dispersion obtained by summing the $(X - \overline{X})$ _____
13. The deviation of a distribution form symmetry is _____
14. In the regression line Y = a+bx , the random value_____
15. What will be the probability of one black card out of 5_2 playing cards_____
16. For the normal distribution will $\mu = 55$ and $\delta = 10$, how much area will be found under curve of thee right of X=55_____
17. If δ^2 is unknown, then we use Z-test if the sample size is_____
18. If internal estimation we always get_____
19. For the degree of freedom V>2 the variance of t-distribution is always____

Subjective

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1. Evaluate (3-5*i*)(-9,-6*i*) (2)

(3-5*i*)(-9,-6*i*)

[3(9)+3(-6*i*) -5*i* (-9) -5*i*(-6*i*)]

[27-18*i*+45*i* -36i^2]

[27+27*i* -36i^2]

2. Define geometric means. (2)

GEOMETRIC MEAN:
The geometric mean, G, of a set of n positive values X1, X2,…,Xn is defined as the positive nth root of their product.

$$G = \sqrt[n]{X_1 X_2 \ldots X_n}$$

(Where $X_i > 0$)
When n is large, the computation of the geometric mean becomes laborious as we have to extract the nth root of the product of all the values.
The arithmetic is simplified by the use of logarithms.

Taking logarithms to the base 10, we get

$$\log G = \frac{1}{n}[\log X_1 + \log X_2 + \ldots + \log X_n]$$

Hence

$$= \frac{\sum \log X}{n}$$

$$G = \text{anti log}\left[\frac{\sum \log X}{n}\right]$$

Example:
Find the geometric mean of numbers:

45, 32, 37, 46, 39,
36, 41, 48, 36.

Solution:
We need to compute the numerical value of

$$= \sqrt[9]{45\times32\times37\times46\times39\times36\times41\times48\times36}$$

But, obviously, it is a bit cumbersome to find the ninth root of a quantity. So we make use of logarithms, as shown below:

X	log X
45	1.6532
32	1.5052
37	1.5682
46	1.6628
39	1.5911
36	1.5563
41	1.6128
48	1.6812
36	1.5563
	14.3870

$$\log G = \frac{\sum \log X}{n}$$

$$= \frac{14.3870}{9} = 1.5986$$

$$\text{Hence } G = \text{anti log } 1.5986$$

$$= 39.68$$

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The above example pertained to the computation of the geometric mean in case of raw data. Next, we consider the computation of the geometric mean in the case of grouped data.

3. For the 10 values we are given $\sum |X_2 - \overline{X}|=20$, find MD (Mean Deviation). (2)

$|d_i| = \sum |X_2 - \overline{X}|=20$; and number of observation **n** is equal to "10"

$$X = \frac{\sum |d_i|}{n}$$

$$X = \frac{20}{10}$$

$$X = 2$$

4. Geometric Sequence?

A sequence, in which each term after the first is found by multiplying the previous term by a constant value called the common ratio, is called a Geometric Sequence (or Geometric Progression).

5. Why t-distribution is a whole family of distributions? (2)

The t-distribution has the following properties:

I. The t-distribution is bell-shaped and symmetric about the value t = 0, ranging from $-\infty$ to ∞.

II. The number of degrees of freedom determines the shape of the t-distribution. Thus there is a different t-distribution for each number of degrees of freedom.

As such, it is a whole family of distributions.

The t-distribution, for small values of **ν**, is flatter than the standard normal distribution which means that the tdistribution is more spread out in the tails than is the standard normal distribution.

6. Density function, $f(x) = \frac{x^2}{4}$ (2)

NORMAL DISTRIBUTION

A continuous random variable is said to be normally distributed with mean μ and standard deviation σ if its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}, \quad -\infty < x < \infty$$

where
$\pi = 3.1416 \simeq 22/7$,
$e \simeq 2.71828$

For any particular value of μ and any particular value of σ, giving different values to x and we obtain a set of ordered pairs (x, f(x)) that yield the bell-shaped curve given above. The formula of the normal distribution defines a *FAMILY* of distributions depending on the values of the two *parameters* μ and σ (as these are the two values that determine the shape of the distribution).

7. Null and Alternative hypothesis. (2)

The null and alternative hypotheses are two mutually exclusive statements about a population. A hypothesis test uses sample data to determine whether to reject the null hypothesis.

Null hypothesis (H_0)

The null hypothesis states that a population parameter (such as the mean, the standard deviation, and so on) is equal to a hypothesized value. The null hypothesis is often an initial claim that is based on previous analyses or specialized knowledge.

Alternative Hypothesis (H_1)

The alternative hypothesis states that a population parameter is smaller, greater, or different than the hypothesized value in the null hypothesis. The alternative hypothesis is what you might believe to be true or hope to prove true.

Step-1:

The null and alternative hypothesis is
H0 : p1 – p2 > 0.10, and
H1 : p1 – p2 < 0.10, where p1 = proportion of educated voters, and p2 = proportion of uneducated voters.

Step-2:

Level of Significance:
$\alpha = 0.05$.

Step-3:

Test Statistic:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0.10}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}$$

which for large sample sizes, is approximately standard normal.

Important Note
In this example, as the hypothesized value of p1 - p2 is not equal to zero, therefore are note estimating the same quantity, and, as such, we do not use in the formula of the test statistic.

Step-4:

Computation:

Here $\hat{p}_1 = \frac{62}{100} = 0.62$, so that $\hat{q}_1 = 0.38$,

$\hat{p}_2 = \frac{69}{150} = 0.46$, so that $\hat{q}_2 = 0.54$.

Thus $$z = \frac{(0.62 - 0.46) - 0.10}{\sqrt{\frac{(0.62)(0.38)}{100} + \frac{(0.46)(0.54)}{150}}}$$

$$= \frac{0.06}{\sqrt{0.002356 + 0.001656}} = \frac{0.06}{0.063} = 0.95.$$

Step-5:
Critical Region:
As this is a one-tailed test, therefore the critical region is given by

$$Z < -z_{0.05} = -1.645$$

Step-6:
Conclusion:
Since the calculated value z = 0.95 does not fall in the critical region, so we accept the null hypothesis
H0 : p1 – p2 > 0.10.The data seems to support the candidate's view.
Until now, we have discussed in considerable detail interval estimation and hypothesis-testing based on the standard normal distribution and the Z-statistic.
Next, we begin the discussion of interval estimation hypothesis-testing based on the t-distribution.

8. If the function *f* is defined by; $f(x)=\frac{3}{x}$, $0< x \leq 3$ (3)

The probability density function of the given uniform distribution is

$f(x)=\frac{3}{\theta}$, $0< x \leq \theta$

$f(x)=\frac{3}{x}$, $0< x \leq 3$

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The first population moment about zero is

$$\mu_1' = \int_0^{\theta} x.\frac{1}{\theta}\,dx = \frac{1}{\theta}\left[\frac{x^2}{s}\right]_0^{\theta} = \frac{\theta}{2}$$

Matching these moments, we obtain:

$$\frac{\sum X_i}{n} = \frac{\theta}{2} = 2\overline{X}$$

Hence, the moment estimator of θ is equal to $2\overline{X}$

$$\overline{\theta} = 2\overline{X}$$

In other words, the moment estimator of θ is just twice the sample mean. It should be noted that, for the above uniform distribution, the mean is given by

$$\mu = \frac{\theta}{2}$$

Solution of the question **$f(x)=\frac{3}{x}$, $0< x \leq 3$**

From the realtion we have devised in above, **take the mean if the maximum value= 3 and multiply it with 2**

$$\overline{X} = \frac{3\times2\times1}{2} = 3$$

$$\overline{\theta} = 2\overline{X}$$

$$\overline{\theta} = 2(3)$$

$$\overline{\theta} = 6$$

9. Find harmonic mean form the following values (10,20,20) (3)

Harmonic mean

In mathematics, the **harmonic mean** (sometimes called the **subcontrary mean**) is one of several kinds of average and in particular one of the Pythagorean means. Typically, it is appropriate for situations when the average of rates is desired.

The harmonic mean is defined as the reciprocal of the arithmetic mean of the reciprocals of the values.

In case of raw data:

$$\text{H.M.} = \frac{n}{\sum(\frac{1}{x})}$$

In case of grouped data (data grouped into a frequency distribution):

$$\text{H.M.} = \frac{n}{\sum f(\frac{1}{x})}$$

10. 62% of students get passed in a particular test. This time 500 students appear in the exam using the normal approximation to the binomial distribution, find the mean and variance of this date. (3)

11. Write down the formulae if combined proportion $\widehat{P_c}$ of two samples. (3)

12. Find P(x=1) , if n=4.5 (3)

13. $f(x) = \frac{1}{5}x$, $f(y) = 2y$ and $f(x,y) = \frac{2}{5}xy$. Prove it x & y are independent. (3)

14. Probability Distribution.

SAMPLING DISTRIBUTION

The probability distribution of any statistic (such as the mean, the standard deviation, the proportion of successes in

a sample, etc.) is known as its sampling distribution. In this regard, the first point to be noted is that there are two ways of sampling --- sampling with replacement, and sampling without replacement. In case of a finite population containing N elements, the total number of possible samples of size n that can be drawn from this population with replacement is Nn. In case of a finite population containing N elements, the total number of possible samples of size n that can be drawn from this population without replacement.

15. How to test Hypothesis?

Hypothesis testing was introduced by Ronald Fisher, Jerzy Neyman, Karl Pearson and Pearson's son, Egon Pearson. Hypothesis testing is a statistical method that is used in making statistical decisions using experimental data. Hypothesis Testing is basically an assumption that we make about the population parameter.

Key terms and concepts:

- ✓ **Null hypothesis:** Null hypothesis is a statistical hypothesis that assumes that the observation is due to a chance factor. Null hypothesis is denoted by; H0: $\mu 1 = \mu 2$, which shows that there is no difference between the two population means.
- ✓ **Alternative hypothesis:** Contrary to the null hypothesis, the alternative hypothesis shows that observations are the result of a real effect.
- ✓ **Level of significance:** Refers to the degree of significance in which we accept or reject the null-hypothesis. 100% accuracy is not possible for accepting or rejecting a hypothesis, so we therefore select a level of significance that is usually 5%.

- Hypothesis-Testing regarding Two Population Means in the Case of Paired Observations (t-distribution)
- The Chi-square Distribution
- Hypothesis Testing and Interval Estimation Regarding a Population Variance (based on Chi-square Distribution)

Error

I. **Type I error:** When we reject the null hypothesis, although that hypothesis was true. Type I error is denoted by alpha. In hypothesis testing, the normal curve that shows the critical region is called the alpha region.

II. **Type II errors:** When we accept the null hypothesis but it is false. Type II errors are denoted by beta. In Hypothesis testing, the normal curve that shows the acceptance region is called the beta region.

Power: Usually known as the probability of correctly accepting the null hypothesis. 1-beta is called power of the analysis.

- ✓ **One-tailed test:** When the given statistical hypothesis is one value like H0: $\mu 1 = \mu 2$, it is called the one-tailed test.
- ✓ **Two-tailed test:** When the given statistics hypothesis assumes a less than or greater than value, it is called the two-tailed test.

Statistical decision for hypothesis testing:

In statistical analysis, we have to make decisions about the hypothesis. These decisions include deciding if we should accept the null hypothesis or if we should reject the null hypothesis. Every test in hypothesis testing produces the significance value for that particular test. In Hypothesis testing, if the significance value of the test is greater than the predetermined significance level, then we accept the null hypothesis. If the significance value is less than the predetermined value, then we should reject the null hypothesis. For example, if we want to see the degree of relationship between two stock prices and the significance value of the correlation coefficient is greater than the predetermined significance level, then we can accept the null hypothesis and conclude that there was no relationship between the two stock prices. However, due to the chance factor, it shows a relationship between the variables.

16. Which variable is more suitable in following examples. (Discrete or Continuous) (5)

i. **The contents of bottle of fluid is Quoted as1025ml.**
ii. **People are asked to state their age at their Birthday.**
iii. **People are asked to state, how many time in the last moth they visited their family doctor.**
iv. **The height of trees at nursery**
v. **The length of time it takes for a light bulb to burn out**

17. If A={2,3,5} , B={1,3,5} , C={2,4,6} Prove that; A∩(B∩C) = (A∩B) ∩C

Left hand side

B∩C = {1,3,5} ∩ {2,4,6}
B∩C = {∅}
A∩(B∩C) = {2,3,5} ∩ {∅}
A∩(B∩C) = {∅}

Right hand side

A∩B = {2,3,5} ∩ {1,3,5}
A∩B = {3,5}
A∩(B∩C) = {2,4,6} ∩ {3,5}
A∩(B∩C) = {∅}

So, **Prove that; A∩(B∩C) = (A∩B) ∩C**

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18. Cartesian method for A = {1,2,3} , B = {a,b}

The Cartesian product of sets A and B, denoted by A × B, (read as "A cross B"), is a set that contains all ordered pairs (x, y), where x belongs to A and y belongs to B. Symbolically, we write **$A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$**

This set is also called the **Cartesian set** of A and B set of A and B, named after the French mathematician Rene' Descartes (1596-1605). The product of a set A by itself is denoted by A2.

This concept of product may be extended to any finite number of sets.

Solution of the **Question;**

A×B = {1,2,3}×{a, b}
A×B = (1, a); (2, a); (3, a); (1, b); (2, b); (3, b)

19. Given $n_1 = n_2 = 16, S_1^2 = 50 , S_2^2 = 16$; Construct a 90% confidence interval of variance ratio δ_1^2 / δ_2^2

20. If, $f(x)$ $0 \le x \le 3$ find range _____ Probability density function

21. Find the 3rd & 4th value from the given values.

If 3rd and 4th terms of geometric series are 12 & B respectively. Find out the sum to infinite.

3rd terms of geometric series

$A_n = a + (n-1)d$
$A_n = 12 + (3-1)12$
$A_n = 12 + (2)12$
$A_n = 12+24$
$A_n = 36$

4rd terms of geometric series

$A_n = a + (n-1)d$
$A_n = 12 + (4-1)12$
$A_n = 12 + (3)12$
$A_n = 12+36$
$A_n = 48$

22. Significance of experiment sampling.

1. Savings in time and money.
   - ✓ Although cost per unit in a sample is greater than in a complete investigation, the total cost will be less (because the sample will be so much smaller than the statistical population from which it has been drawn)

- ✓ A sample survey can be completed faster than a full investigation so that variations from sample unit to sample unit over time will largely be eliminated.
- ✓ Also, the results can be processed and analyzed with increased speed and precision because there are fewer of them.

2. More detailed information may be obtained from each sample unit.
3. Possibility of follow-up:
   (After detailed checking, queries and omissions can be followed up --- a procedure which might prove impossible in a complete survey).
4. Sampling is the only feasible possibility where tests to destruction are undertaken or where the population is effectively infinite. The next two important concepts that need to be considered are those of sampling and non-sampling errors.

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23. If $Z = 3+7i$ find $|Z.Z^*|$

$Z=3-7i$ $Z^*=3+7i$

Then,

$Z.Z^* = (3-7i)(3+7i)$

$Z.Z^* = [3(3)+3(7i)-7i(3)-7i(7i)]$

$Z.Z^* = [9+21i-21i-49i^2]$

$Z.Z^* = [4-49(-1)]$

$Z.Z^* = [4+49]$

$Z.Z^* = [53]$

24. Find mean deviation?

The mean deviation can also be viewed in another way:

It is the arithmetic mean of the deviations of the first and third quartiles round the median i.e.

$$\frac{(M - Q_1) + (Q_3 - M)}{2}$$

$$\frac{M - Q_1 + Q_3 - M}{2}$$

$$\frac{Q_1 + Q_3}{2}$$

25. Difference between Ratio and internal dota.

A **set in mathematics** is a collection of well defined and distinct objects, considered as an object in its own right.

26. What is variation? And definition of geometric mean.

The Arithmetic, Geometric and Harmonic means are averages that are mathematical in character, and give an indication of the magnitude of the observed values.

The **geometric mean**, G, of a set of n positive values X1, X2,…,Xn is defined as the positive nth root of their product.

$G = \sqrt[n]{X_1 X_2 \ldots . X_n}$

Taking logarithms to the base 10, we get

$$\log G = \frac{1}{n}[\log X_1 + \log X_2 + ... + \log X_n]$$

Hence

$$= \frac{\sum \log X}{n}$$

Example:

$$G = \text{anti log}\left[\frac{\sum \log X}{n}\right]$$

27. Explain inferential Statistic

Inferential statistics makes inferences about populations using data drawn from the population. Instead of using the entire population to gather the data, the statistician will collect a sample or samples from the millions of residents and make inferences about the entire population using the sample.

28. Draw the Skewed curve.

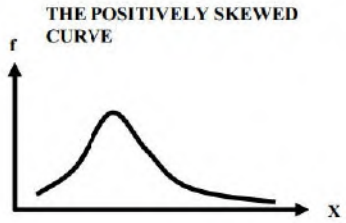
In a **skewed curve**, the median and mean are not the same, as is the case with a bell **curve**. There could be **two curves**

I. **Positive skewed curves**

II. **Negative skewed curves**

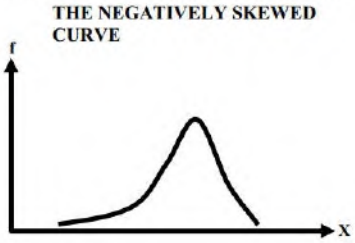
Positive skewed curves

The positively skewed curve is that one whose right tail is longer than its left tail, as shown below



Negative skewed curves

The negatively skewed frequency curve is the one for which the left tail is longer than the right tail



29. Example of power and super set.

POWER SET

The class of ALL subsets of a set A is called the Power Set of A and is denoted by P(A).

For example, if A = {H, T}, then P(A) = {ϕ, {H}, {T}, {H, T}}.

SUPER SET

A set AA is a superset of another set BB if all elements of the set BB are elements of the set AA. The superset relationship is denoted as A⊃BA⊃B.

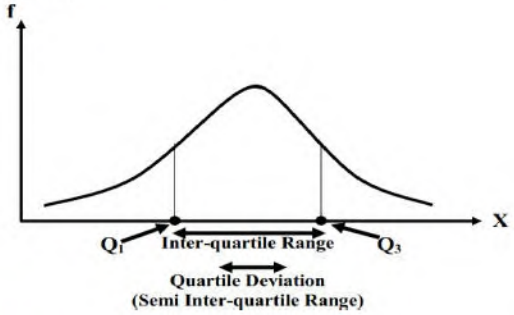
For example, if AA is the set {♢,♡,♣,♠}{♢,♡,♣,♠} and BB is the set {♢,♣,♠}{♢,♣,♠}, then A⊃BA⊃B but B⊃/AB⊅A. Since AA contains elements not in BB, we can say that AA is a proper superset of BB. Or if I1I1 is the interval [0,2][0,2] and I2I2 is the interval [0,1][0,1], then I1⊃I2I1⊃I2.

30. Given Q_1 & Q_3 , find the Quartile range.

The quartile deviation is defined as half of the difference between the third and first quartiles i.e.

$$Q.C = \frac{Q_3 - Q_1}{2}$$

It is also known as **semi-interquartile range.**



31. Write regression equation for y on x and also find intercept a.

On numerous occasions, our interest lies not in just one single variable but in two, three, four or more variables. For example, if we talk about the yield of a crop, we realize that the yield of any crop depends on a variety of factors ---- the fertility of the soil, the type of fertilizer used, the amount of rainfall, and so on.

- ✓ Simple Linear Regression
- ✓ Standard Error of Estimate
- ✓ Correlation

Simple Linear Regression $Y = a + bX$

32. Average heights of school is 5.2 inch 12 students height are given. Calculate "sampling error".

The difference between the estimate derived from the sample (i.e. the statistic) and the true population value (i.e. the parameter) is technically called the sampling error.

Sampling error $= \overline{X} - \mu$

For example,

Sampling error arises due to the fact that a sample cannot exactly represent the pop, even if it is drawn in a correct manner

Solution of the *Question*

Sampling error = $\overline{X} - \mu$
Sampling error = $5.2 - 12$
Sampling error = -6.8

33. Write two properties of X^2 distribution

When we are reasonably sure that the shape of the observed frequency distribution is quite similar to that of the normal distribution. (As indicated in the case of the fitting of the binomial distribution to real data), in order to decide whether or not our fitted normal distribution is a reasonably good fit, the proper statistical procedure is the **Chi-square Test of Goodness of Fit**. Hypothesis Testing and Interval Estimation Regarding a Population Variance is based on Chi square Distribution.

Mathematical equation

The mathematical equation of the Chi-Square distribution is as follows:

$$f(x) = \frac{1}{2^{v/2}\, r\left(\frac{v}{2}\right)} (x)^{\left(\frac{v}{2}\right)-1} . e^{-x/2}, \mathbf{0} < x \infty$$

$$f(x) = \frac{1}{2^{v/2}\Gamma(v/2)} (x)^{(v/2)-1} . e^{-x/2}, \quad 0 < x < \infty$$

This distribution has only one parameter ν, which is known as the degrees of freedom of the Chi-Square distribution.

34. Standard normal variable.

The **standard normal distribution** is a special case of the normal distribution. It is the distribution that occurs when a normal random variable has a mean of zero and a standard deviation of one.

Standard Score (aka, z-score)

The normal random variable of a standard normal distribution is called a **standard score** or a **z-score**. Every normal random variable X can be transformed into a z score via the following equation:

$$z = (X - \mu) / \sigma$$

Where X is a normal random variable, μ is the mean of X, and σ is the standard deviation of X.

35. Four most common types' uses of variables.

A "**variable**" in algebra really just means one thing—an unknown value. However, in statistics, you'll come across dozens of types of variables in statistics. In most cases, the word still means that you're dealing with something that's unknown, but—unlike in algebra—that unknown isn't always a number. Some variable types are used more than others. For example, you'll be much more likely to come across continuous variables than you would dummy variables.

Click on any bold variable name to learn more about that particular type.

- **Categorical variable**: variables than can be put into categories. For example, the category "Toothpaste Brands" might contain the variables Colgate and Aquafresh.
- **Confounding variable**: extra variables that have a hidden effect on your experimental results.
- **Continuous variable**: a variable with infinite number of values, like "time" or "weight".
- **Control variable**: a factor in an experiment which must be held constant. For example, in an experiment to determine whether light makes plants grow faster, you would have to control for soil quality and water.
- **Dependent variable**: the outcome of an experiment. As you change the independent variable, you watch what happens to the dependent variable.
- **Discrete variable**: a variable that can only take on a certain number of values. For example, "number of cars in a parking lot" is discrete because a car park can only hold so many cars.
- **Independent variable**: a variable that is not affected by anything that you, the researcher, does. Usually plotted on the x-axis
- **Qualitative variable**: a broad category for any variable that can't be counted (i.e. has no numerical value). Nominal and ordinal variables fall under this umbrella term.

- **Quantitative variable:** A broad category that includes any variable that can be counted, or has a numerical value associated with it. Examples of variables that fall into this category include discrete variables and ratio variables.

36. Four most commonly used types of averages.

The most common types of averages are:

I. The arithmetic mean
II. The geometric mean
III. The harmonic mean
IV. The median
V. The mode

37. Suppose that X = {a,b}. Possible combination of X and find value of C (2,2)

38. In normal distribution, how much data full in this interval $\mu \pm \delta^2$

39. Find stem and leaf of the data 55,51,62,91,64,66,75,73,71,82.

A **stem** is the leading digit(s) of each number and is used in sorting, while a **leaf** is the rest of the number or the trailing digit(s) and shown in display. A vertical line separates the leaf (or leaves) from the stem.

***Solution of the* Question**

Stem of the data	Leaf of the data
5	1,5
6	2,4
7	1,3,5

40. Find domain and range of $f(x) = 5x+2$

The set consisting of members of the pre-image or inputs of a function is called its **domain**. For a given domain the set of possible outcomes or images of a function is called its **range**.

$f(x) = 5x+2$; Domain: $x = R$, Range: $f(x) = R$

41. A teacher is making multiple choice quizzes. She wants students same question but question appears in different order. If there are 27 students what least number does quiz contain.

27! =

42. Define Qualitative and quantitative data, also give example.

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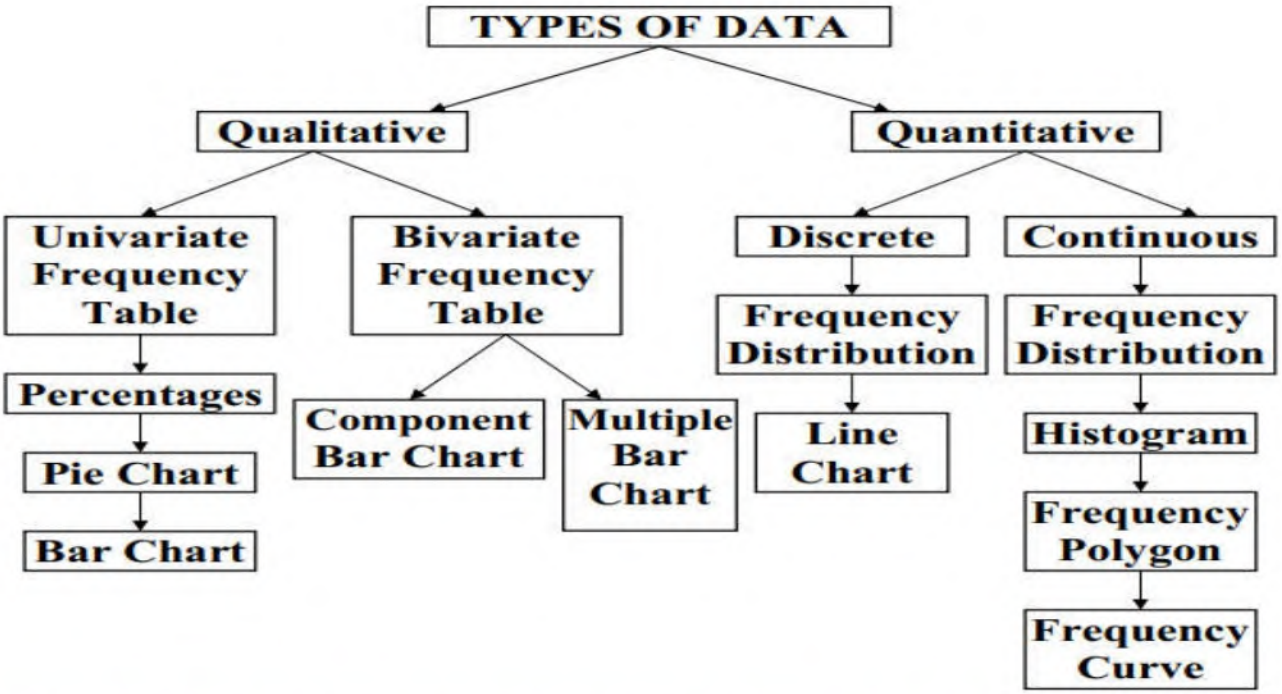
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43. Write formula sample is drawn from a population mean 33 variance δ^2 if S = 15 $\overline{X}$ = 40.4. Then *Compute* t-statics

44. Following are the two random sample.

Sample 1	$N_1 = 8$	$\overline{X_1} = 5.17$	$\sum x^2 = 810$
Sample 2	$N_1 = 9$	$\overline{X_2} = 2.8$	$\sum x^2 = 600$

Construct 95% confidential interval for the difference between mean (where *Syx* = 9.75)

45. Completing Square method; $x^2+10x+5 = 0$

It can't be solved. As we can make factor of 10*x*.

46. Explain; $(\frac{\delta^2}{2})(\frac{a^2}{V-2})$

Thank You !

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FINAL TERM EXAMINATION

STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
For a particular data the value of Pearson's coefficient of skewness is greater then zero. What will be the shape of distribution?
► Negatively skewed
► J-shaped
► Symmetrical
► **Positively skewed**

Question No: 2 (Marks: 1) - Please choose one
In measures of relative dispersion unit of measurement is:
► Changed
► **Vanish**
► Does not changed
► Dependent

Question No: 3 (Marks: 1) - Please choose one
The F-distribution always ranges from:
► 0 to 1
► 0 to -∞
► -∞ to +∞
► **0 to +∞**

Question No: 4 (Marks: 1) - Please choose one
In chi-square test of independence the degrees of freedom are:
► **n - p**
► n - p-1
► n - p- 2
► n – 2

Question No: 5 (Marks: 1) - Please choose one
The Chi- Square distribution is continuous distribution ranging from:
► $-\infty \leq \chi 2 \leq \infty$
► $-\infty \leq \chi 2 \leq 1$
► $-\infty \leq \chi 2 \leq 0$
► **$0 \leq \chi 2 \leq \infty$**

Question No: 6 (Marks: 1) - Please choose one
If X and Y are random variables, then $E(X-Y)$ is equal to:

- $E(X)+E(Y)$
- $E(X)-E(Y)$
- $X-E(Y)$
- $E(X)-Y$

Question No: 7 (Marks: 1) - Please choose one
If ŷ is the predicted value for a given x-value and b is the y-intercept then the equation of a regression line for an independent variable x and a dependent variable y is:

► **$\hat{y} = mx + b$, where m = slope**
► $x = \hat{y} + mb$, where m = slope
► $\hat{y} = x/m + b$, where m = slope
► $\hat{y} = x + mb$, where m = slope

Question No: 8 (Marks: 1) - Please choose one
The location of the critical region depends upon:
► Null hypothesis
► **Alternative hypothesis**
► Value of alpha
► Value of test-statistic

Question No: 1 (Marks: 1) - Please choose one
The t-Distribution is**……** Spread out then the standard normal Distribution.
► Less
►**More**
►Equally
►Not

Question No: 2 (Marks: 1) - Please choose one
To find the confidence internal for the ratio of two variances we use:
►**F-Distribution**
►Z-Distribution
►Chi-Square Distribution
►T-Distribution

Question No: 3 (Marks: 1) - Please choose one
How many percent of values are less then 4th deciles in a symmetric distribution.
►14
►24
►4
►**40**

Question No: 4 (Marks: 1) - Please choose one
The combined distribution of more than two random variables is:
►Bivariate Distribution
►Marginal Distribution
►**Joint Distribution**
►Univariate Distribution

Question No: 5 (Marks: 1) - Please choose one
The degrees of freedom for a T-test with sample size 14 is:
►14
►**13**
►**7**
►0

Question No: 6 (Marks: 1) - Please choose one
Which of the falling is true for the binomial distribution b(x: n, p):
►**Mean > Variance**
►Mean < Variance
►Mean = Variance
►Mean = Standard Deviation

Question No: 7 (Marks: 1) - Please choose one
What is mf in the formula of mode?
►first frequency
►last frequency
►middle frequency
►**highest frequency**

Question No: 1 (Marks: 1) - Please choose one
The parameters of the binomial distribution b(x; n, p) are:

- x & n
- x & p
- **n & p**
- x, n & p

Question No: 2 (Marks: 1) - Please choose one
Which of the following is true for the Poisson distribution:

- mean > variance
- mean < variance
- **mean = variance**
- mean= standard deviation

Question No: 3 (Marks: 1) - Please choose one
If a significance level of 1% is used rather than 5%, the null hypothesis is:

- More likely to be rejected
- **Less likely to be rejected**
- Just as likely to be rejected
- None of the above

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Question No: 4 (Marks: 1) - Please choose one
The variance of the chi-square distribution is:

- $2v$
- $v-1$
- $v-2$
- v

Question No: 5 (Marks: 1) - Please choose one
The degrees of freedom for a t-test with sample size 10 is:

- 5
- 8
- **9**
- 10

Question No: 1 (Marks: 1) - Please choose one
The value of χ^2 can never be :

- Zero–
- Less than 1
- Greater than 1
- **Negative**

Question No: 2 (Marks: 1) - Please choose one
The mean of the F-distribution is:

- $\frac{v_1}{v_1-2} \quad for v_1 \rangle 2$
- $\frac{v_2}{v_2-2} \quad for v_2 \rangle 2$
- $\frac{v_1}{v_1-2} \quad for v_1 \geq 2$
- $\frac{v_2}{v_2-2} \quad for v_1 \leq 2$

Question No: 3 (Marks: 1) - Please choose one

The F-distribution always ranges from:

- ► 0 to 1
- ► 0 to -∞
- ► -∞ to +∞
- ► **0 to +∞**

Question No: 4 (Marks: 1) - Please choose one

The total number of samples when sampling is done with replacement :

- ► N^n
- ► C_n^N
- ► $N-n$
- ► 1_{N-1}

Question No: 5 (Marks: 1) - Please choose one

ANOVA was introduced by :

- ► Helmert
- ► Pearson
- ► **R.A Fisher**
- ► Francis

Question No: 6 (Marks: 1) - Please choose one

The test statistic used in analysis of variance procedure follow the ..... distribution.:

- ► χ^2
- ► T
- ► Z
- ► **F**

Question No: 7 (Marks: 1) - Please choose one

For testing of hypothesis about population proportion , we use:

- ► **Z-test**
- ► t-Test
- ► Both Z & T-test
- ► F test

Question No: 8 (Marks: 1) - Please choose one

A die is rolled. What is the probability that the number rolled is greater than 2 and even:

- ► 1/2
- ► **1/3 2/6 = 1/3**
- ► 2/3
- ► 5/6

Hint: - number that is greater than 2 and is even = 4,6 i.e only 2 numbers out of 6
Therefore the probability 2/6 = 1/3

Question No: 10 (Marks: 1) - Please choose one

The probability of drawing a king of spade from a pack of 52 cards is:

- ► 1/4
- ► 1/13
- ► 1/26
- ► **1/52**

Question No: 11 (Marks: 1) - Please choose one

An estimator T is said to be unbiased estimator of θ if

- ► **E (T) =** θ
- ► E (T) =T
- ► E (T) =0
- ► E (T) =1

-

Question No: 12 (Marks: 1) Please choose one

From point estimation, we always get:

- ► **Single value**
- ► Two values
- ► Range of values
- ► Zero

Question No: 13 (Marks: 1) - Please choose one

The best unbiased estimator for population variance is:

- ► Sample mean
- ► Sample median
- ► Sample proportion
- ► **Sample variance**

Question No: 14 (Marks: 1) - Please choose one

When c is a constant, then E(c) is:

- ► 0
- ► 1
- ► **c**
- ► -c

Question No: 15 (Marks: 1) - Please choose one

Var(4X + 5) =_____

- ► 16 Var (X)
- ► **16 Var (X) + 5 (correct)**
- ► 4 Var (X) + 5
- ► 12 Var (X)

Question No: 16 (Marks: 1) - Please choose one

When f(x) is continuous probability function, then P(X = 1) is:

- ► 1
- ► ∞
- ► $-\infty$
- ► **0 (Page 188)**

Question No: 17 (Marks: 1) - Please choose one

The hyper geometric random variable is a(an):

- ► Continuous variable
- ► **Discrete variable**
- ► Undefined
- ► Independent variable

Question No: 18 (Marks: 1) - Please choose one

From a sample of 200 people were asked whether they like a particular product. Fifty said 'yes' and remain said 'no', assuming 'yes' means a success, which of the following is correct?

- ► Sample proportion p=0.33
- ► **Sample proportion p=0.25 (Page 245)**
- ► Population proportion p= 0.33
- ► Population proportion p=0.25

Question No: 19 (Marks: 1) - Please choose one

In any data set, what percent of values fall in the interval $Median \pm Q.D$?

- ► **50 per cent**
- ► 68.5 per cent
- ► 95.4 per cent
- ► 99 per cent

Question No: 20 (Marks: 1) - Please choose one

If $\sum_{i=1}^{5}(X_i - 20) = 0, then\ X = ...$

► **0**
► 20
► 5
► 25

Question No: 21 (Marks: 1) - Please choose one

The height of a student is 60 inches. This is an example of .....?

► **Continuous data (correct)**
► Qualitative data
► Categorical data
► Discrete data

Question No: 22 (Marks: 1) - Please choose one

In Statistics, we have MSE which is abbreviation of……

► **Mean square error**
► Measured square error
► Medical screening exam
► Major sampling error

Question No: 23 (Marks: 1) - Please choose one

Which one is the formula of mid range:

► $x_m - x_0$

► $x_0 - x_m$

► $\frac{x_0 - x_m}{2}$

► $\frac{x_0 + x_m}{2}$

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Question No: 24 (Marks: 1) - Please choose one

The deviation of a distribution from symmetry is called:

► Kurtosis
► **Skewness**
► Dispersion
► Flatness

Question No: 25 (Marks: 1) - Please choose one

If E is an impossible event, then P(E) is:

► 1
► 2
► **0**
► 0.5

Question No: 26 (Marks: 1) - Please choose one

If a data set has the even number of observations, the median :

► Is the average value of the two middle items
► Can not be determined
► must be equal to the mean
► **Is the average value of the two middle items when all items are arranged in ascending order**

Question No: 27 (Marks: 1) - Please choose one

For the Poisson distribution $P(X=1)=\frac{e^{-0.135}0.135^1}{1!}$ the mean value is :

- ► 2
- ► 5
- ►10
- ► **0.135**

Question No: 28 (Marks: 1) - Please choose one

In testing of hypothesis, we always begin it with assuming that:

- ► **Null hypothesis is true**
- ► Alternative hypothesis is true
- ► Sample size is large
- ► Population is normal

Question No: 29 (Marks: 1) - Please choose one

Variance of the t-distribution is given by the formula:

- ► $\sigma^2 = \frac{\nu}{\nu-2}$
- ► $\sigma^2 = \frac{\nu^2}{\nu-2}$
- ► $\sigma^2 = \frac{\nu}{\nu-1}$
- ► $\sigma^2 = \frac{\nu}{\nu-2}$

Question No: 30 (Marks: 1) - Please choose one

If a continuous probability distribution has $\beta_2 = 2.14$ then what will be peakedness of the distribution?

- ► **Platykurtic**
- ► Mesokurtic
- ► Leptokutic
- ► Moderately skewed

Question No: 1 (Marks: 1) - Please choose one

When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

- ► Mutually exclusive
- ► **Equally likely**
- ► Not mutually exclusive
- ► Exhaustive

Question No: 3 (Marks: 1) - Please choose one

The LSD test is applied only if the null hypothesis is:

- ► **Rejected**
- ► Accepted
- ► No conclusion
- ► Acknowledged

Question No: 4 (Marks: 1) - Please choose one

Analysis of variance is a procedure that enables us to test the equality of several:

- ► Variances
- ► **Means**
- ► Proportions
- ► Groups

Question No: 5 (Marks: 1) - Please choose one

ANOVA was introduced by :

- ► Helmert
- ► Pearson
- ► **R.A Fisher**
- ► Francis

Question No: 6 (Marks: 1) - Please choose one

For testing of hypothesis about population proportion , we use:

- ► **Z-test**
- ► t-Test
- ► Both Z & T-test
- ► F test

Question No: 7 (Marks: 1) - Please choose one

If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

- ► **0,1,2,3**
- ► 1,3,3,1
- ► 1, 2, 3
- ► 3, 2

Question No: 8 (Marks: 1) - Please choose one

If X and Y are independent variables, then E (XY) is:

- ► E(XX)
- ► **E(X).E(Y)**
- ► X.E(Y)
- ► Y.E(X)

Question No: 9 (Marks: 1) - Please choose one

The parameters of the binomial distribution b(x; n, p) are:

- ► x & n
- ► x & p
- ► **n & p**
- ► x, n & p

Question No: 10 (Marks: 1) - Please choose one

If P (E) is the probability that an event will occur, which of the following must be false:

- ► **P(E)= - 1**
- ► P(E)=1
- ► P(E)=1/2
- ► P(E)=1/3

Question No: 11 (Marks: 1) - Please choose one

An estimator T is said to be unbiased estimator of if

- ► **E (T) = σ^2**
- ► E (T) =T
- ► E (T) =0
- ► E (T) =1

Question No: 12 (Marks: 1) - Please choose one

The best unbiased estimator for population variance is:

- ► **Sample mean**
- ► Sample median
- ► Sample proportion
- ► Sample variance

Question No: 13 (Marks: 1) - Please choose one

The sample variance $S^2 = \Sigma(x-x)^2$ is:

- ► Unbiased estimator of n
- ► **Biased estimator of** μ
- ► Unbiased estimator of μ
- ► None of these

Question No: 14 (Marks: 1) - Please choose one

When c is a constant, then E(c) is:

- ► 0
- ► 1
- ► **c**
- ► -c

Question No: 15 (Marks: 1) - Please choose one
If f (x, y) is bivariate probability density function of continuous r.v.'s X and Y then

$g(x)$ is:

- $\int_{-\infty}^{\infty} f(x,y)\,dx$
- $\int_{-\infty}^{\infty} f(x,y)\,dy$
- $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)dx\,dy$ **(Page 199)**
- $\int_{a}^{b}\int_{c}^{d} f(x,y)\,dy\,dx$
-

Question No: 16 (Marks: 1) - Please choose one
The analysis of variance technique is a method for :

- Comparing F distributions
- **Comparing three or more means (Page 320)**
- Measuring sampling error
- Comparing variances

Question No: 17 (Marks: 1) - Please choose one
The continuity correction factor is used when:

- The sample size is at least 5
- Both *nP* and *n (1-P)* are at least 30
- **A continuous distribution is used to approximate a discrete distribution** Click here for detail
- The standard normal distribution is applied

Question No: 18 (Marks: 1) - Please choose one
Stem and leaf is more informative when data is :

- Equal to 100
- Greater Than 100
- **Less than 100** click here for detail
- In all situations

Question No: 19 (Marks: 1) - Please choose one
The branch of Statistics that is concerned with the procedures and methodology for obtaining valid conclusions is called:

- ► Descriptive Statistics
- ► Advance Statistics
- ► **Inferential Statistics (Page 237)**
- ► Sampled Statistics

Question No: 20 (Marks: 1) - Please choose one
Which of the following is a systematic arrangement of data into rows and columns?

- ► Classification
- ► **Tabulation**
- ► Bar chart
- ► Component bar chart

Question No: 21 (Marks: 1) - Please choose one
In normal distribution Q.D =

- ► 0.5σ
- ► 0.75σ
- ► 0.7979σ
- ► 0.6745σ Click here for detail

Question No: 22 (Marks: 1) - Please choose one
In normal distribution $\beta_2 =$

- ► 1
- ► 2
- ► **3 (Page 119)**
- ► 0

Question No: 23 (Marks: 1) - Please choose one

If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- ► Ogive
- ► Frequency polygon
- ► **Frequency curve (Page 38)**
- ► Historigram

Question No: 24 (Marks: 1) - Please choose one
Which one of the following statements is true regarding a population?

► **It must be a large number of values (Page 12)**
► It must refer to people
► It is a collection of individuals, objects, or measurements
► It is small part of whole

Question No: 25 (Marks: 1) - Please choose one
When $Q_1 = 2$ *and* $Q_3 = 4$,what is the value of Median, if the distribution is symmetrical:

► 1
► **2**
► 3
► 4

Question No: 26 (Marks: 1) - Please choose one
In a simple linear regression model, if it is assumed that the intercept parameter is equal to zero, then:

► The regression line will pass through the origin
► The regression line will pass through the point (0,10).
► The regression line will pass through the point (0,-10).
► **The slope of the line will also be equal to 0.**

Question No: 27 (Marks: 1) - Please choose one
The degrees of freedom for a t-test with sample size 10 is:

► 5
► 8
► **9 (Page 298) rep**
► 10

Question No: 28 (Marks: 1) - Please choose one
In testing of hypothesis, we always begin it with assuming that:

► **Null hypothesis is true (Page 277) rep**
► Alternative hypothesis is true
► Sample size is large
► Population is normal

Question No: 29 (Marks: 1) - Please choose one
A failing student is passed by an examiner is an example of:

- ► Type I error
- ► **Type II error**
- ► Correct decision
- ► No information regarding student exams

Question No: 30 (Marks: 1) - Please choose one
How to find $P(X+Y \leq 1)$?

- ► f(0, 0) + f(0, 1) + f(1, 2)
- ► f(2, 0) + f(0, 1) + f(1, 0)
- ► f(0, 0) + f(1, 1) + f(1, 0)
- ► f(0, 0) + f(0, 1) + f(1, 0)

FINAL TERM EXAMINATION

STA301- Statistics and Probability (Session - 3)

Question No: 1 (Marks: 1) - Please choose one
The total number of samples when sampling is done with replacement :

- ► **(Page 237) rep**
- ►
- ►
- ► 1

Question No: 2 (Marks: 1) - Please choose one
The test statistic used in analysis of variance procedure follow the ..... distribution.:

- ► χ^2
- ► T
- ► Z
- ► **F (Page 326) rep**

Question No: 3 (Marks: 1) - Please choose one
If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

- ► **0,1,2,3 rep**
- ► 1,3,3,1
- ► 1, 2, 3
- ► 3, 2

Question No: 4 (Marks: 1) - Please choose one
If X and Y are independent variables, then E (XY) is:

- ► E(XX)
- ► **E(X).E(Y) (Page 202) rep**
- ► X.E(Y)
- ► Y.E(X)

Question No: 5 (Marks: 1) - Please choose one

The sample variance is:

- ► Unbiased estimator of
- ► **Biased estimator of** **(Page 260) rep**
- ► Unbiased estimator of
- ► None of these

Question No: 6 (Marks: 1) - Please choose one
When c is a constant, then E(c) is:

► 0
► 1
► **c (Page 180) rep**
► -c

Question No: 7 (Marks: 1) - Please choose one
When the random variable X and Y are independent, its co-variance is:

► One
► Negative
► **Zero (Page 207)**
► Positive

Question No: 8 (Marks: 1) - Please choose one
When f (x, y) is bivariate probability density function of continuous r.v.'s X and Y, then

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$$

is equal to:

► **1 (Page 199)**
► 0
► -1
► ∞

Question No: 9 (Marks: 1) - Please choose one
Which dispersion is calculated from all the observations?

► **Standard deviation**
► Quartile deviation
► Rang
► Coefficient of Rang

Question No: 10 (Marks: 1) - Please choose one
Standard deviation of the data 7, 7, 7, 7, 7, 7, 7 is

► 49
► Sqrt (7)
► **0 Standard deviation will always be zero if all the values in data are same**
► 7

Question No: 11 (Marks: 1) - Please choose one
Which one is the poor measure of dispersion in open-end distribution?

- ► **Range**
- ► Standard deviation
- ► Mean deviation
- ► Variance

Question No: 12 (Marks: 1) - Please choose one
Men tend to marry women who are slightly younger than themselves. Suppose that every man married a woman who was exactly 5 years younger than themselves. Which of the following is correct:

- ► The correlation is −5
- ► The correlation is 5
- ► **The correlation is 1** Click here for detail
- ► The correlation is 0

Question No: 13 (Marks: 1) - Please choose one
Sum of absolute deviations of the values is least when deviations are taken from:

- ► **Mean**
- ► Median
- ► Mode
- ► Geometric mean

Question No: 14 (Marks: 1) - Please choose one
Which of the following measures of central location is affected most by extreme values:

- ► Geometric Mean
- ► Median
- ► **Mean** Click here for detail
- ► Mode

Question No: 15 (Marks: 1) - Please choose one
Which of the following is a critical value of Z when $1-\alpha = 95\%$ for one tailed test:

- ► 2.58
- ► 1.645
- ► 1.25
- ► **1.96**

Question No: 16 (Marks: 1) - Please choose one
In normal distribution Q.D =

- 0.5σ
- 0.75σ
- 0.7979σ
- 0.6745σ Click here for detail rep

Question No: 17 (Marks: 1) - Please choose one
The difference between expected value of statistic and parameter is called:

- Non-sampling error
- Sampling error
- Standard error
- **Bias** Click here for detail

Question No: 18 (Marks: 1) - Please choose one
In Statistics, we have MSE which is abbreviation of……

- **Mean square error (Page 330) rep**
- Measured square error
- Medical screening exam
- Major sampling error

Question No: 19 (Marks: 1) - Please choose one
The following data shows the number of hours worked by 200 statistics students.

Number of Hours	Frequency
0 - 9	40
10 - 19	50
20 - 29	70
30 - 39	40

What is its class interval?

- 9
- **10**
- 11
- Varies from class to class

$x_0 - x_m$

Question No: 20 (Marks: 1) - Please choose one

Which one is the formula of range:

$\frac{x_0 - x_m}{2}$

$\frac{x_0 + x_m}{2}$

(Page 80)

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Question No: 21 (Marks: 1) - Please choose one

Which one is the formula of Geometric mean for group data:

$anti\log\left[\frac{\sum f \log X}{n}\right]$

$anti\log\left[\frac{\sum \log X}{n}\right]$ (Page 75)

$anti\log\left[\sum \log f X\right]$

$anti\log\left[\frac{\sum \log f X}{\sum_{i=1}^{n} f}\right]$

Question No: 22 (Marks: 1) - Please choose one

The F-distribution has …… parameter.

- One
- No
- **Two (Page 312)**
- Three

Question No: 23 (Marks: 1) - Please choose one

The sample mean $X = \frac{\Sigma X}{n}$ is an unbiased estimator of population mean (μ), because:

- $E(X) = 0$
- $E(X) = \mu$ **(Page 259)**
- $E(X) > \mu$
- $E(X) < \mu$

Question No: 24 (Marks: 1) - Please choose one

For a particular hypothesis test, $\alpha = 0.05$ and $\beta = 0.10$. The power of test equals to:

- 0.95
- 0.25
- **0.90**
- 0.14

Question No: 25 (Marks: 1) - Please choose one

The degrees of freedom for a t-test with sample size 10 is:

- 5
- 8
- **9 (Page 298) rep**
- 10

Question No: 26 (Marks: 1) - Please choose one

The degrees of freedom for a t-test with sample size 14 is:

- 14
- **13 (Page 341) rep**
- 7
- 0

Question No: 27 (Marks: 1) - Please choose one

The degrees of freedom for a t-test with sample size 6 is:

- 1
- 3
- **5 (Page 341)**
- 7

Question No: 28 (Marks: 1) - Please choose one
In testing of hypothesis, we always begin it with assuming that:

- ► **Null hypothesis is true (Page 277) rep**
- ► Alternative hypothesis is true
- ► Sample size is large
- ► Population is normal

Question No: 29 (Marks: 1) - Please choose one

If a continuous probability distribution has then what will be peakedness of the distribution?

- ► **Platykurtic (Page 119) rep**
- ► Mesokurtic
- ► Leptokutic
- ► Moderately skewed

Question No: 30 (Marks: 1) - Please choose one
The difference between statistic and parameter is called:

- ► **Bias** Click here for detail **rep**
- ► Standard error
- ► Sampling error
- ► None of above

FINAL TERM EXAMINATION

STA301- Statistics and Probability (Session - 4)

Question No: 1 (Marks: 1) - Please choose one
Mean deviation is always:

- ► Less than S.D
- ► Greater than S.D
- ► Greater or equal to S.D
- ► **Less or equal to S.D** Click here for detail

Question No: 2 (Marks: 1) - Please choose one
The value of χ^2 can never be :

- ► Zero
- ► Less than 1
- ► Greater than 1
- ► **Negative (Page 307) rep**

Question No: 3 (Marks: 1) - Please choose one
The mean of the F-distribution is:

- ►
- **(Page 312) rep**
- ►
- ►
- ►

Question No: 4 (Marks: 1) - Please choose one

If X and Y are random variables, then is equal to:

- ►
- **(Page 202) rep**
- ►
- ►
- ►

Question No: 5 (Marks: 1) - Please choose one
Evaluate: (9-4)!

- ► 362880
- ► **120**
- ► 24
- ► 6

Question No: 6 (Marks: 1) - Please choose one
Which formula represents the probability of the complement of event A:

- ► $1 + P(A)$
- ► **$1 - P(A)$ (Page 156)**
- ► $P(A)$
- ► $P(A)$ -1

Question No: 7 (Marks: 1) - Please choose one
Ideally the width of confidence interval should be:

- ► **0 (Page 270)**
- ► 1
- ► 99
- ► 100

Question No: 8 (Marks: 1) - Please choose one
If the sampling distribution of X is normal, the interval $\mu_x \pm 3\sigma_x$ includes:

- ► 99% of the sample means
- ► **99.73% of the sample means (Page 228)**
- ► 98% of the sample means
- ► 95% of the sample means

Question No: 9 (Marks: 1) - Please choose one
The probability distribution of a statistic is called the:

- ► Population distribution
- ► Frequency distribution
- ► **Sampling distribution** click here for detail
- ► Sample distribution

Question No: 10 (Marks: 1) - Please choose one

An estimator T is said to be unbiased estimator of if

- ► **E (T) = (Page 258) rep**
- ► E (T) =T
- ► E (T) =0
- ► E (T) =1

Question No: 11 (Marks: 1) - Please choose one

If the following is a probability distribution, then what is the value of 'a':

X	1	2	3
P(X)	0.1	a	0.1

- ► 0.6
- ► **0.8**
- ► 0.2
- ► 0.4

Question No: 12 (Marks: 1) - Please choose one

A discrete probability function f(x) is always:

- ► Non-negative
- ► Negative
- ► **One (Page 168)**
- ► Zero

Question No: 13 (Marks: 1) - Please choose one

An expected value of a random variable is equal to:

- ► Variance
- ► **Mean (Page 191)**
- ► Standard deviation
- ► Covariance

Question No: 14 (Marks: 1) - Please choose one

$f(x|1) =$

The _____:

- $f(1,1)$
- $f(x,1)$
- $f(x,1)$ $h(1)$ **(Page 198)**
- $f(x,1)$ $h(x)$

Question No: 15 (Marks: 1) - Please choose one

The area under a normal curve between 0 and -1.75 is

- .0401
- .5500
- **.4599 (Page 230)**
- .9599

Question No: 16 (Marks: 1) - Please choose one

The continuity correction factor is used when:

- The sample size is at least 5
- Both *nP* and *n (1-P)* are at least 30
- **A continuous distribution is used to approximate a discrete distribution** Click here for detail
- The standard normal distribution is applied

Question No: 17 (Marks: 1) - Please choose one

Which of the following is impossible in sampling:

- Destructive tests
- **Heterogeneous**
- To make voters list
- None of these

Question No: 18 (Marks: 1) - Please choose one
Which of the following is a systematic arrangement of data into rows and columns?

- ► Classification
- ► **Tabulation rep**
- ► Bar chart
- ► Component bar chart

Question No: 19 (Marks: 1) - Please choose one
Which one of the following statements is true regarding a sample?

- ► **It is a part of population (Page 13)**
- ► It must contain at least five observations
- ► It refers to descriptive statistics
- ► It produces True value

Question No: 20 (Marks: 1) - Please choose one
The data for an ogive is found in which distribution?

- ► A relative frequency distribution
- ► A frequency distribution
- ► A joint frequency distribution
- ► **A cumulative frequency distribution (Page 43)**

FINAL TERM EXAMINATION

STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
10! =……

- ► 362880
- ► **3628800**
- ► 362280
- ► 362800

Question No: 2 (Marks: 1) - Please choose one
When E is an impossible event, then P(E) is:

► 2
► **0 (Page 146) rep**
► 0.5
► 1

Question No: 3 (Marks: 1) - Please choose one
The value of χ^2 can never be :
► Zero
► Less than 1
► Greater than 1
► **Negative (Page 307) rep**

Question No: 4 (Marks: 1) - Please choose one
The curve of the F- distribution depends upon:
► **Degrees of freedom (Page 312)**
► Sample size
► Mean
► Variance

Question No: 5 (Marks: 1) - Please choose one

If X and Y are random variables, then is equal to:

►
► **(Page 202) rep**
►
►

Question No: 6 (Marks: 1) - Please choose one
In testing hypothesis, we always begin it with assuming that:

► **Null hypothesis is true (Page 277) rep**
► Alternative hypothesis is true
► Sample size is large
► Population is normal

Question No: 7 (Marks: 1) - Please choose one

$$\frac{e^{-0.135}\ 0.135^{1}}{1!}$$

For the Poisson distribution P(x) = the mean value is :

►2
► 5
► 10
► 0.135 (Page 222) rep

Question No: 8 (Marks: 1) - Please choose one
When two coins are tossed simultaneously, P (one head) is:

1
► 4
1
► 2
3
► 4
► 1

Question No: 9 (Marks: 1) - Please choose one
From point estimation, we always get:

► Single value (Page 257) rep
► Two values
► Range of values
► Zero

Question No: 10 (Marks: 1) - Please choose one

The sample variance is:

► Unbiased estimator of
► Biased estimator of **(Page 260) rep**
► Unbiased estimator of
► None of these

Question No: 11 (Marks: 1) - Please choose one
Var(4X + 5) =_____

► 16 Var (X)
► **16 Var (X) + 5 rep**
► 4 Var (X) + 5
► 12 Var (X)

Question No: 12 (Marks: 1) - Please choose one
When f (x, y) is bivariate probability density function of continuous r.v.'s X and Y, then

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$$

is equal to:

► **1 rep**
► 0
► -1
► ∞

Question No: 13 (Marks: 1) - Please choose one
The area under a normal curve between 0 and -1.75 is

► .0401
► .5500
► **.4599 (Page 230) rep**
► .9599

Question No: 14 (Marks: 1) - Please choose one
When a fair die is rolled, the sample space consists of:
► 2 outcomes
► 6 outcomes
► **36 outcomes (Page 145)**
►16 outcomes

Question No: 15 (Marks: 1) - Please choose one
When testing for independence in a contingency table with 3 rows and 4 columns, there are _____ degrees of freedom.
► 5
► **6 (Page 341)**
► 7
► 12

Question No: 16 (Marks: 1) - Please choose one
The F- test statistic in one-way ANOVA is:

- ► SSW / SSE
- ► MSW / MSE
- ► SSE / SSW
- ► **MSE / MSW (Not sure)**

Question No: 17 (Marks: 1) - Please choose one
The continuity correction factor is used when:

- ► The sample size is at least 5
- ► Both *nP* and *n (1-P)* are at least 30
- ► **A continuous distribution is used to approximate a discrete distribution rep**
- ► The standard normal distribution is applied

Question No: 18 (Marks: 1) - Please choose one
A uniform distribution is defined by:

- ► **Its largest and smallest value** click here for detail
- ► Smallest value
- ► Largest value
- ► Mid value

Question No: 19 (Marks: 1) - Please choose one
Which graph is made by plotting the mid point and frequencies?

- ► **Frequency polygon (Page 34)**
- ► Ogive
- ► Histogram
- ► Frequency curve

Question No: 20 (Marks: 1) - Please choose one
In a set of 20 values all the values are 10, what is the value of median?

- ► 2
- ► 5
- ► **10**
- ► 20

FINAL TERM EXAMINATION

STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
For a particular data the value of Pearson's coefficient of skewness is greater then zero. What will be the shape of distribution?

► **Negatively skewed**
► J-shaped
► Symmetrical
► **Positively skewed (Page 110)**

Question No: 2 (Marks: 1) - Please choose one
In measures of relative dispersion unit of measurement is:

► **Changed**
► **Vanish Rep**
► Does not changed
► Dependent

Question No: 3 (Marks: 1) - Please choose one
The F-distribution always ranges from:

► 0 to 1
► 0 to -∞
► -∞ to +∞
► **0 to +∞ (Page 312) rep**

Question No: 4 (Marks: 1) - Please choose one
In chi-square test of independence the degrees of freedom are:

► **n - p**
► n - p-1
► n - p- 2
► n – 2

Question No: 5 (Marks: 1) - Please choose one
The Chi- Square distribution is continuous distribution ranging from:

- ► $-\infty \leq \chi^2 \leq \infty$
- ► $-\infty \leq \chi^2 \leq 1$
- ► $-\infty \leq \chi^2 \leq 0$
- ► **$0 \leq \chi^2 \leq \infty$ (Page 307) rep**

Question No: 6 (Marks: 1) - Please choose one

If X and Y are random variables, then is equal to:

- ►
- ► **(Page 202) rep**
- ►
- ►

Question No: 7 (Marks: 1) - Please choose one
If $\hat{y}$ is the predicted value for a given x-value and b is the y-intercept then the equation of a regression line for an independent variable x and a dependent variable y is:

- ► **$\hat{y} = mx + b$, where m = slope (Page 121) rep**
- ► $x = \hat{y} + mb$, where m = slope
- ► $\hat{y} = x/m + b$, where m = slope
- ► $\hat{y} = x + mb$, where m = slope

Question No: 8 (Marks: 1) - Please choose one
The location of the critical region depends upon:

- ► **Null hypothesis**
- ► **Alternative hypothesis (Page 281) rep**
- ► Value of alpha
- ► Value of test-statistic

Question No: 9 (Marks: 1) - Please choose one
The variance of the t-distribution is give by the formula:

▶

▶

▶

(Page 293)

▶

Question No: 10 (Marks: 1) - Please choose one
Which one is the correct formula for finding desired sample size?

$$n = \left(\frac{Z_{\alpha_2} . \sigma}{e} \right)^2$$ **(Page 276)**

▶

$$n = \left(\frac{Z_{\alpha_2} . \ \sigma}{e} \right)^2$$

▶

$$n = \left(\frac{Z_{\alpha_2} . X}{e} \right)^2$$

▶

$$n = \frac{Z_{\alpha_2} . \sigma}{e}$$

▶

Question No: 11 (Marks: 1) - Please choose one
A discrete probability function f(x) is always:

- ► Non-negative
- ► Negative
- ► **One (Page 168) rep**
- ► Zero

Question No: 12 (Marks: 1) - Please choose one
E(4X + 5) =_____

- ► 12 E (X)
- ► **4 E (X) + 5**
- ► 16 E (X) + 5
- ► 16 E (X)

Question No: 13 (Marks: 1) - Please choose one
How P(X + Y < 1) can be find:

- ► f(0, 0) + f(0, 1) + f(1, 2)
- ► f(2, 0) + f(0, 1) + f(1, 0)
- ► f(0, 0) + f(1, 1) + f(1, 0)
- ► f(0, 0) + f(0, 1) + f(1, 0)

Question No: 14 (Marks: 1) - Please choose one
$f(x|1) =$
The _____:

- ► $f(1,1)$
- ► $f(x,1)$
- ► $\frac{f(x,1)}{h(1)}$ **(Page 198) rep**
- ► $\frac{f(x,1)}{h(x)}$

Question No: 15 (Marks: 1) - Please choose one
The area under a normal curve between 0 and -1.75 is

- .0401
- .5500
- **.4599 (Page 230) rep**
- .9599

Question No: 16 (Marks: 1) - Please choose one
In normal distribution M.D. =

- 0.5σ
- 0.75σ
- 0.7979σ Click here for detail **rep**
- 0.6445σ

Question No: 17 (Marks: 1) - Please choose one
In an ANOVA test there are 5 observations in each of three treatments. The degrees of freedom in the numerator and denominator respectively are.....

- 2, 4
- 3, 15
- 3, 12
- **2, 12**

Question No: 18 (Marks: 1) - Please choose one
A set that contains all possible outcomes of a system is known as

- Finite Set
- Infinite Set
- **Universal Set (Page 134)**
- **No of these**

Question No: 19 (Marks: 1) - Please choose one
Stem and leaf is more informative when data is :

- Equal to 100
- Greater Than 100
- **Less than 100** click here for detail rep
- In all situations

Question No: 20 (Marks: 1) - Please choose one
A population that can be defined as the aggregate of all the conceivable ways in which a specified event can happen is known as:

- ► Infinite population
- ► Finite population
- ► Concrete population
- ► **Hypothetical population (Page 12)**

STA 301 – Quizzes

Question # 1 of 10 (Total Marks: 1) Select correct option:
The number of telephone calls that pass through a switchboard has a Poison distribution with mean equal to 2 per minute. The probability that no telephone calls pass through the switchboard in two consecutive minutes is:

0.2707
0.0517
0.0183
0.0366

Question # 2 of 10 (Total Marks: 1) Select correct option:
The range of the binomial distribution is:
0, 1, 2, ... , 100
0, 1, 2, … , n Click here for detail
0, 1, 2, … , x
1, 2, … , n

Question # 3 of 10 (Total Marks: 1) Select correct option:
Which of the following is NOT CORRECT about a standard normal distribution?

$P(0 \leq Z \leq 1.50) = .4332$
$P(Z \geq 2.0) = .0228$
$P(Z \geq -2.5) = .4938$ (Page 230)
$P(Z \leq -1.0) = .1587$

Question # 4 of 10 (Total Marks: 1) Select correct option:
The distribution function (df) is also known as cumulative distribution function (cdf).
Yes (Page 173)
No

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following pairs of events are mutually exclusive?

A:the numbers above100;B: the numbers less than-200
A:the odd numbers; B:the number 5
A:the even numbers; B:the numbers greater than 10
A:the numbers less than 5; B:all the negative numbers

Question # 6 of 10 (Total Marks: 1) Select correct option:
Two events A & B are said to be independent if...
P(A) + P(B)
P(B/A) = P(B)
P(A) * P(B) (Page 162)
P(A/B) = P(A)

Question # 7 of 10 (Total Marks: 1) Select correct option:
The collection of all outcomes for an experiment is called:

A sample space Click here for detail
Joint probability simple event
The intersection of events
Random experiment

Question # 8 of 10 (Total Marks: 1) Select correct option:
Symbolically, a conditional probability is:
P(AB)
P(A/B) (Page 159)
P(A)
P(AUB)

Question # 9 of 10 (Total Marks: 1) Select correct option:
Which of the following statements is INCORRECT about the sampling distribution of the sample mean?

The standard error of the sample mean will decrease as the sample size increases
The standard error of the sample mean is measure of the variability of the sample mean among repeated samples
The sample mean is unbiased for the true (unknown) population mean
The sampling distribution shows how the sample was distributed around the sample mean
Click here for detail

Question # 10 of 10 (Total Marks: 1) Select correct option:
If one event is unaffected by the outcome of another event the two events are said to be:

1. Dependent
2. Not Mutually Exclusive
3. Mutually Exclusive

Independent Click here for detail

Question # 1 of 10 (Total Marks: 1) Select correct option:
Let X be a random variable with binomial distribution, that is (X=0,1,…, n). The expected value E[X] is

p
np (Page 214)
np(1-p)
Xnp

Question # 2 of 10 (Total Marks: 1) Select correct option:
The sample mean is an unbiased estimator for the population mean. This means:

The sample mean has a normal distribution
The average sample mean, over all possible samples, equals the population mean (Page 259)
The sample mean is always very close to the population mean
The sample mean will only vary a little from the population mean

Question # 4 of 10 (Total Marks: 1) Select correct option:
Probability of an impossible event is always:

Less than one
Greater than one
Between one and zero
Zero (Page 146)

Question # 5 of 10 (Total Marks: 1) Select correct option:
The function abbreviated to d.f. is also called the.....

Probability density function
Probability distribution function
Commutative distribution function (Page 173)
Discrete function

Question # 6 of 10 (Total Marks: 1) Select correct option:
The total area under the normal curve is:

0
1 (Page 186)
0.5
0.75

Question # 7 of 10 (Total Marks: 1) Select correct option:
Two events A & B are said to be independent if....

P (A) + P (B)
P (B\A) = P (B)
P (A) * P (B) (Page 162) rep
P (A\B) = P (A)

Question # 8 of 10 (Total Marks: 1) Select correct option:
When two coins are tossed the probability of at most one head is:

1/4
2/4
3/4 rep
1

Question # 9 of 10 (Total Marks: 1) Select correct option:
For exhaustive events, the P(AUBUC) is equal to:

P(A)
P(S) (Page 147)
P(A) * P(B)* P(C)
P(B)

Question # 10 of 10 (Total Marks: 1) Select correct option:
One card is drawn from a standard 52 card deck. In describing the occurrence of two possible events, an Ace and a King, these two events are said to be:
Select correct option:
independent
randomly independent
random variables
mutually exclusive click here for detail

STA 301 – Quizzes

Question # 1 of 10 (Total Marks: 1) Select correct option:
The number of parameters in hypergeometric distribution is (are):

1
2
3 (Page 219)
4

Question # 2 of 10 (Total Marks: 1) Select correct option:
If Y=bX, then variance of Y is

b*2 var(x)
var(x)
b var(x)
b square root var(x)

Question # 3 of 10 (Total Marks: 1) Select correct option:
If f(x) is a continuous probability function, then P(X = 2) is:

1
0 (Page 186)
1/2
2

Question # 4 of 10 (Total Marks: 1) Select correct option:
In regression line Y=a+bX, Y is called:

Dependent variable (Page 121)
Independent variable
Explanatory variable
Regressor

Question # 5 of 10 (Total Marks: 1) Select correct option:
If A and B are mutually exclusive events with P (A) =0.25 and P (B) = 0.50, Then P (A or B) =.....

- 0.25
- 0.75 (Page 154)
- 0.50
- 1

Question # 6 of 10 (Total Marks: 1) Select correct option:
In a 52 well shuffled pack of 52 playing cards, the probability of drawing any one diamond card is

- 1/52
- 4/52
- 13/52
- 52/52

Question # 7 of 10 (Total Marks: 1) Select correct option:
Probability of a sure event is

- 8
- 1 (Page 154)
- 0
- 0.5

Question # 8 of 10 (Total Marks: 1) Select correct option:
If Y=3X+5,then S.D of Y is equal to

- 9 s.d(x)
- 3 s.d(x)
- s.d(x)+5
- 3s.d(x)+5

Question # 9 of 10 (Total Marks: 1) Select correct option:
The probability of drawing a red queen card from well-shuffled pack of 52 playing cards is

- 4/52
- 2/52
- 13/52
- 26/52

Question # 10 of 10 (Total Marks: 1) Select correct option:
If P (B|A) = 0.25 and P (A and B) =0.20, then P (A) is

0.05
0.80 (Page 159)
0.95
0.75

Question # 1 of 10 (Total Marks: 1) Select correct option:
A student solved 25 questions from first 50 questions of a book to be solved. The probability that he will solve the remaining all questions is:

0.25
0.5
1
0

Question # 1 of 10 (Total Marks: 1) Select correct option:
The classical definition of probability assumes:

Exhaustive events
Mutually exclusive events
Equally likely evens (Page 151)
Independent evens

Question # 2 of 10 (Total Marks: 1) Select correct option:
In scatter diagram, the variable plotted along Y-axis is:
Independent variable
Dependent variable
Continuous variable
Discrete variable

Question # 3 of 10 (Total Marks: 1) Select correct option:
Which of the following measures of dispersion are based on deviations from the mean?

Variance
Standard deviation
Mean deviation
All of the these

Question # 4 of 10 (Total Marks: 1) Select correct option:
What does it mean when a data set has a standard deviation equal to zero?
All values of the data appear with the same frequency.
The mean of the data is also zero.
All of the data have the same value. Click here for detail
There are no data to begin with.

Question # 5 of 10 (Total Marks: 1) Select correct option:
A set of possible values that a random variable can assume and their associated probabilities of occurrence are referred to as _____.

Probability distribution
The expected return
The standard deviation
Coefficient of variation

Question # 6 of 10 (Total Marks: 1) Select correct option:
Which of the following can never be probability of an event?

0
1
0.5
-0.5

Question # 7 of 10 (Total Marks: 1) Select correct option:
The standard deviation of -1, -1, -1, -1 will be

1
-1
0
Does not exist

Question # 8 of 10 (Total Marks: 1) Select correct option:
The Special Rule of Addition is used to combine:

Independent Events
Mutually Exclusive Events click here for detail
Events that total more than 1.00
Events based on subjective probabilities

Question # 9 of 10 (Total Marks: 1) Select correct option:
set which is the sub-set of every set is

Empty Set (Page 134)
Power Set
Universal Set
Super Set

Question # 10 of 10 (Total Marks: 1) Select correct option:
When two dice are rolled the number of possible sample points is :

6
12
24
36 (Page 145)

Question # 1 of 10 (Total Marks: 1) Select correct option:
If two events A and B are not mutually exclusive then

P (A or B) = P (A) + P (B) – P (A and B) (Page 157)
P (A or B) = P (A) + P (B)
P (A or B) = P (A) x P (B)
P (A or B) = P (A) + P (B)

Question # 2 of 10 (Total Marks: 1) Select correct option:
Evaluate (10-4)!

1000
720
480
32

Question # 3 of 10 (Total Marks: 1) Select correct option:
When we toss a coin , we get only:

1 outcome
2 outcome
3 outcome
4 outcome

Question # 4 of 10 (Total Marks: 1) Select correct option:
If we roll a die then probability of getting a '6' will be

2/6
1/6
4/6
1

Question # 5 of 10 (Total Marks: 1) Select correct option:
If P(A) = 0.45, P(B) = 0.35, and P(A and B) = 0.25, then P(A | B) is:

1.4
1.8
0.714 0.25/0.35 = 0.714 (Page 159)
0.556

Question # 6 of 10 (Total Marks: 1) Select correct option:
Which of the following is not a measure of central tendency?

Percentile
Quartile
Standard deviation
Mode

Question # 7 of 10 (Total Marks: 1) Select correct option:
Random experiment can be repeated any no. of times under the…… conditions.
Different
Similar (Page 144)

Question # 9 of 10 (Total Marks: 1) Select correct option:
The simultaneous occurrence of two events is called:
Joint probability (Page 194)
Subjective probability
Prior probability
Conditional probability

Question # 10 of 10 (Total Marks: 1) Select correct option:
In regression analysis, the variable that is being predicted is the

Dependent variable click here for detail
Independent variable
Intervening variable
None of these

Question # 1 of 10 (Total Marks: 1) Select correct option:
The probability of continuous random variable x on any particular point is always zero..
Yes (Page 186)
No

Question # 2 of 10 (Total Marks: 1) Select correct option:
P(an event) =no of favorable outcome/total no. of outcomes is the definition of
Subjective approach (Page 149)
Objective approach

Question # 3 of 10 (Total Marks: 1) Select correct option:
If C is a constant ,then E(c)=
0
1
C (Page 180) rep
-c

Question # 4 of 10 (Total Marks: 1) Select correct option:
When we toss a fair coin 4 times, the sample space consists of….points.
4
8
12
16

Question # 5 of 10 (Total Marks: 1) Select correct option:
If we roll three fair dices then the total number of outcomes is:
6
36
216 6*6*6=216
1296

Question # 6 of 10 (Total Marks: 1) Select correct option:
The probability of an event is always:
greater than 0
less than 1
between o and 1 click here for detail
greater than 1

Question # 7 of 10 (Total Marks: 1) Select correct option:
In a multiplication theorem P (A and B) equals:
P(A)
P (B) P(A) + P (B)
P(A) * P (B|A) (Page 159)
P(B\A)*P(B)

Question # 8 of 10 (Total Marks: 1) Select correct option:
If a die is rolled, what is the probability of getting an even number greater than 2?
1/2
1/3 2/6 = 1/3
2/3
5/6

Question # 9 of 10 (Total Marks: 1) Select correct option:
In a Discrete probability distribution, P (x > 23) is read as:
P (there are more than 23 successes)
P (there are less than 23 successes)
P (there are at least 23 successes)
P (there are at most 23 successes)

Question # 10 of 10 (Total Marks: 1) Select correct option:
A dormitory on campus houses 200 students. 120 are male, 50 are upper division students, and 40 are upper division male students.A student is selected at random. The probability of selecting a lower division student, given the student is a female, is:

7/8
7/20
7/15
1/4

Question # 1 of 10 (Total Marks: 1) Select correct option:
The function F(x) gives the probability of the event that X takes a value .....
Less than x
Greater or equal to x
Less or equal x (Page 173)
Equal to x

Question # 2 of 10 (Total Marks: 1) Select correct option:
In a simple regression line model ,it is assume that the intercept parameter is equal to zero,
The regression line will pass through the origin.
The regression line will pass through the point (0,10)
The regression line will pass through the point (0,-10)
The slope of the line will also be zero.

Question # 3 of 10 (Total Marks: 1) Select correct option:
If p(AnB)=p(A/B).p(B),then A and B are
Independent
Dependant
Equally likely
Mutually exclusively

Question # 4 of 10 (Total Marks: 1) Select correct option:
A fair coin is tossed three times, the probability that at least one head appears,

1/8
7/8 (HHH,HHT,HTH, HTT, THH THT TTH TTT)
3/8
5/8

Question # 5 of 10 (Total Marks: 1) Select correct option:
In probability distribution, the sum of probabilities is equals to

0
0.1
0.5
1 click here for detail

Question # 6 of 10 (Total Marks: 1) Select correct option:
When a coin is tossed 3 times, the probability of getting 3 tails is

1/8 (HHH,HHT,HTH, HTT, THH THT TTH TTT)
3/8
3/6
2/8

Question # 7 of 10 (Total Marks: 1) Select correct option:
In how many ways can a team of 11 players be chosen from a total of 16 players?

4368
2426
5400
2680

Question # 8 of 10 (Total Marks: 1) Select correct option:
The standard deviation of c (constant) is

c
c square
0 click here for detail
does not exist

Question # 9 of 10 (Total Marks: 1) Select correct option:
Let E and F be events associated with the same experiment. Suppose the E and F are independent and that P(E) = 1/4 and P(F) = 1/2 Then P(E U F) is:

1/8
3/4 (Page 157)
7/8
5/8

STA 301 – Quizzes

Question # 1 of 10 (Total Marks: 1) Select correct option:
In which of the following situations binomial distributions is approximate to normal distribution?

n = 50, p = 0.01
n = 500, p = 0.001
n = 100, p = 0.05
n = 50, p = 0.02 click here for detail

Question # 2 of 10 (Total Marks: 1) Select correct option:
The location and shape of the normal curve is (are) determined by:

Mean
Variance
Mean & variance
Mean & standard deviation click here for detail

Question # 3 of 10 (Total Marks: 1) Select correct option:
A random experiment has five outcomes in its sample space {s1, s2, s3, s4, s5}. If P(s1)=0.2,P(s2)=0.3, P(s3)=0.1 and P(s4)=0.2 then P(s5)=?

1
0.2
0.8
0.5

Question # 4 of 10 (Total Marks: 1) Select correct option:
The joint density function f(x,y) will be a pdf if

Both of integrals f(x,y) dx dy =1 (Page 200)
Both of integrals f(x,y) dx dy >1
Both of integrals f(x,y) dx dy <1
Both of integrals f(x,y) dx dy =0

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following is correct property for joint probability distribution of X and Y:

Sigma f(X,Y)=1 (Page 194)
Sigma f(Y,X)=1
Both of above
None of above

Question # 6 of 10 (Total Marks: 1) Select correct option:
A random variable that can assume every possible value in an interval [a, b]:

Discrete variable
Continuous variable (Page 9)
Qualitative variable
Categorical variable

Question # 7 of 10 (Total Marks: 1) Select correct option:
Normal approximation to the binomial distribution is used when:

np>5
nq>5
Both of the above (Page 235)
None of the above

Question # 8 of 10 (Total Marks: 1) Select correct option:
The Maximum Likelihood Estimators (MLE) are …… and …… but not necessarily ……
Unbiased, consistent, efficient
Consistent, unbiased, efficient
Unbiased, efficient, consistent
Consistent, efficient, unbiased (Page 264)

Question # 9 of 10 (Total Marks: 1) Select correct option:
A probability density function 'f(x)' has the following property:

f(x) <= 0
f(x) < 0
f(x) > 0 (Page 186)
f(x) >= 0

Question # 10 of 10 (Total Marks: 1) Select correct option:
For a continuous random variable X, P(X = x) is:
Select correct option:

0 (Page 186)
0.5
1
undefined

Question # 1 of 10 (Total Marks: 1) Select correct option:
An urn contains 4 red balls and 6 green balls. A sample of 4 balls is selected from the urn without replacement. It is the example of:

Binomial distribution
Hypergrometric distribution (Page 219)
Poisson distribution
Exponential distribution

Question # 2 of 10 (Total Marks: 1) Select correct option:
The probability distribution of the proportion of successes in all possible samples is called the:

Sampling distribution (Page 237)
Sampling probability distribution
Sampling distribution of sample proportions
Sampling distribution of Population proportions

Question # 3 of 10 (Total Marks: 1) Select correct option:
For good approximation of Poisson distribution to the binomial distribution, which of the following condition (s) is/are required:

The population size is large relative to the sample size
The probability, p, is close to .5 and the population size is large
The probability, p, is small and the sample size is large (Page 222)
The probability, p, is close to .5 and the sample size is large

Question # 4 of 10 (Total Marks: 1) Select correct option:
If an estimator is more efficient then the other estimator, its shape of the sampling distribution will be

Flattered
Normal
Highly peaked (Page 261)
Skewed to right

Question # 6 of 10 (Total Marks: 1) Select correct option:
Match the binomial probability P(x< 23) with the correct statement.

P(there are at most 23 successes)
P (there are fewer than 23 successes)
P (there are more than 23 successes)
P (there are at least 23 successes)

Question # 7 of 10 (Total Marks: 1) Select correct option:
In statistics, the term 'expected value' implies the ___ value.

Independent
Normal
Standard
Mean (Page 191)

Question # 8 of 10 (Total Marks: 1) Select correct option:
A quantity obtained by applying certain rule or formula is known as

Estimate (Page 264)
Estimator
Parameter
Proportion

Question # 9 of 10 (Total Marks: 1) Select correct option:
Total no. of possible samples of size 2 (without replacement) from the population of size 6, will be:

20
15
10
18

Question # 10 of 10 (Total Marks: 1) Select correct option:
When a coin is tossed 3 times, the probability of getting 3 or less tails is

1/2 (not sure)
0
1
3/5

Question # 1 of 10 (Total Marks: 1) Select correct option:
Total no. of possible samples of size 3 (with replacement) from the population of size 6, will be:

256
196
216 6^3
325

Question # 2 of 10 (Total Marks: 1) Select correct option:
Using the normal approximation to the binomial distribution with n= 3 and p= 0.0571 the value of mean is:

0.1713 (Page 235)
0.2132
0.5133
0.1923

Question # 3 of 10 (Total Marks: 1) Select correct option:
Suppose 60% of a herd of cattle is infected with a particular disease. Let Y = the number of non-diseased cattle in a sample of size 5. the distribution of Y is:

Binomial with n = 5 and p = 0.6
Binomial with n = 5 and p = 0.4
Binomial with n = 5 and p = 0.5
Poisson with u = .6

Question # 4 of 10 (Total Marks: 1) Select correct option:
The probability of success changes from trial to trial, is the property of:

Binomial experiment (Page 211)
Hypergeometric experiment
Both binomial & hypergeometric experiment
Poisson experiment

Question # 5 of 10 (Total Marks: 1) Select correct option:
If a statistic used as an estimator, has its expected value equal to the true value of the population parameter being estimated then it is called.....

Consistent
Unbiased (Page 258)
Efficient
Sufficient

Question # 6 of 10 (Total Marks: 1) Select correct option:
In moments method, how many equations are needed to find the 2 unknown parameters?

2 (Page 263)
3
n/2

Question # 7 of 10 (Total Marks: 1) Select correct option:
Which of the following is desirable for a good point estimator?

Consistency
Unbiasedness
Efficiency
All of these (Page 258)

Question # 8 of 10 (Total Marks: 1) Select correct option:
The distribution function (df) is also known as

Probability distribution
Probability mass function
Probability density function
Cumulative distribution function (Page 173)

Question # 9 of 10 (Total Marks: 1) Select correct option:
If X and Y are two discrete r.v's with joint probability function f(x,y), then the conditional probability function Y given X, f(y|x) is given by

f(yj | xi) = f(xi, yj) / g(xi) (Page 195)
f(yj | xi) = f(xi, yj) / h(yj)
f(yj | xi) = f(xi, yj) / Sum of g(xi)
f(yj | xi) = f(xi, yj) / sum of h(yj)

Question # 1 of 10 (Total Marks: 1) Select correct option:
Which of the probability distributions has three parameters?

Binomial distribution
Normal distribution
Hypergeometric distribution (Page 219)
Poisson distribution

Question # 2 of 10 (Total Marks: 1) Select correct option:
A standard deviation obtained from sampling distribution of sample statistics is known as

Sampling error
Standard error (Page 240)
Minimum error
Universal error

Question # 3 of 10 (Total Marks: 1) Select correct option:
A parameter is a ..... quantity.

Constant
Variable
Sample
Random

Question # 4 of 10 (Total Marks: 1) Select correct option:
How can you define statistical inference?

A decision, estimate, prediction or generalization about the population based on information contained in a sample click here for detail

A statement made about a sample based on the measurements in that sample
A set of data selected from a larger set of data
A decision, estimate, prediction or generalization about sample based on information contained in a population

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following is most important and most widely used method in point estimation?

The method of moments
The method of fractional moments
The method of leas square (Page 263)
The method of maximum likelihood

Question # 6 of 10 (Total Marks: 1) Select correct option:
Binomial distribution is skewed to the right if:

p=q
P<q (Page 215)
p>q
p=n

Question # 7 of 10 (Total Marks: 1) Select correct option:
In a discrete distribution function, F(23) can be stated as:

P (there are at most 23 successes)
P (there are at least 23 successes)
P (there are less than 23 successes)
P (there are more than 23 successes)

Question # 8 of 10 (Total Marks: 1) Select correct option:
The Probabilities of the various values of the sample satistic can be computed using the..... definition of probability.

Subjective
Objective
Classical (Page 149)
None of the baove

Question # 9 of 10 (Total Marks: 1) Select correct option:
E(10X +3) =_____

10 E(X)
E(X)+3
10 E(X)+3
100E(X)

Question # 10 of 10 (Total Marks: 1) Select correct option:
If p is very small and n is considerably large then we shall apply the:

Binomial distribution
Hypergeometric distribution
Poisson distribution (Page 222)
Exponential distribution

Question # 1 of 10 (Total Marks: 1) Select correct option:
Which of the following is NOT applicable to a Poisson distribution?

IF P = 0.5 & n = 19
IF P = 0.01 & n = 200
IF P = 0.02 & n = 300
IF P = 0.03 & n = 500

Question # 2 of 10 (Total Marks: 1) Select correct option:
The normal distribution has points of infection which are equidistance from the:

Median
Mean (Page 229)
Mode
Mean,Median & Mode

Question # 3 of 10 (Total Marks: 1) Select correct option:
If a random variable X denotes the number of heads when we toss a fair coin 5 times, the X assumed the values:

0,1,2,3
1, 2,3,4,5
0, 1, 2,3,4,5
1, 5, 5

Question # 4 of 10 (Total Marks: 1) Select correct option:
As a rule of thumb, when n>=30, then we can assume that……is normally distributed:

Probability distribution
Sampling distribution (Page 243)
Binomial distribution
Sampling distribution of sample mean

Question # 5 of 10 (Total Marks: 1) Select correct option:
If b(x, 7, 0.30), the variance of this distribution is:

1.77
1.74
1.44
1.47 (Page 333)

Question # 6 of 10 (Total Marks: 1) Select correct option:
Which of the following is a characteristic of a binomial probability experiment?

Each trial has more than two possible outcomes
P(success) = P(failure)
Probability of success changes for each trail
The result of one trial does not affect the probability of success on any other trial (Page 211)

Question # 7 of 10 (Total Marks: 1) Select correct option:
The number of parameters in uniform distribution is (are):

1
2 (Page 225)
3
4

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Question # 8 of 10 (Total Marks: 1) Select correct option:
The probability can never be:

1
1/2
1
-1/2

Question # 9 of 10 (Total Marks: 1) Select correct option:
The conditional probability P (A\B) is:

P(A n B)/P(B) (Page 159)
P(A n B)/P(A)
P(A U B)/P(B)
P(A U B)/P(A)

Question # 10 of 10 (Total Marks: 1) Select correct option:
A random sample of n=25 values gives sample mean 83. Can this sample be regarded as drawn from a normal population with μ= 80 and s= 7? In this question the alternative hypothesis will be:

H1: μ = 80
H1: μ ≠ 80
H1: μ > 80
H1: μ <80

Question # 1 of 10 (Total Marks: 1) Select correct option:
The binomial distribution is negatively skewed when:
p>q (Page 215)
p<q
p=q
p=q=1/2

Question # 2 of 10 (Total Marks: 1) Select correct option:
When we draw the sample with replacement, the probability distribution to be used is:
Binomial
Hypergeometric
Binomial & hypergeometric
Poisson

Question # 3 of 10 (Total Marks: 1) Select correct option:
The moment ratios of normal distribution come out to be:
0 and 1
0 and 2
0 and 3 (Page 227)
0 and 4

Question # 4 of 10 (Total Marks: 1) Select correct option:
Suppose the test scores of 600 students are normally distributed with a mean of 76 and standard deviation of 8. The number of students scoring between 70 and 82 is:
272
164
260
328 click here for detail

Question # 5 of 10 (Total Marks: 1) Select correct option:
If P (A) = 0.3 and P (B) = 0.5, find P (A/B) where 'A' and 'B' are independent:

0.3
0.5
0.8
0.15 (Page 162)

Question # 6 of 10 (Total Marks: 1) Select correct option:
If the second moment ratio is less than 3 the distribution will be:
Mesokurtic
Leptokurtic
Platykurtic
None of these

Question # 7 of 10 (Total Marks: 1) Select correct option:
For the independent events A and B if P (A) = 0.25, P (B) =0.40 then P (A and B) =……
Select correct option:

0.65
0.1 (Page 162)
0.50
0.15

Question # 8 of 10 (Total Marks: 1) Select correct option:
A random variable X has a probability distribution as follows: X | 0 1 2 3 P(X) | 2k 3k 13k 2k What is the possible value of k:

0.01
0.03
0.05
0.07

Question # 9 of 10 (Total Marks: 1) Select correct option:
The probability of drawing any one spade card is:
1/52
4/52
13/52
52/52

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